
1. OBJECTIVE

Analyze Instagram posts and engagement data to identify best posting times, content types with high engagement and follower growth signals for Alfido Tech's social media strategy.

2. DATASET INFORMATION

Source : Kaggle - Instagram Dataset

Link : kaggle.com/datasets/bhanupratapbiswas/instagram

Tools : Python, Pandas, NumPy, Matplotlib, Seaborn, WordCloud

3. DATA CLEANING STEPS

- Renamed columns for easier processing
- Converted timestamp to datetime format
- Extracted hour, day and month from timestamp
- Handled missing values in all columns
- Computed engagement metrics per post

4. KEY FINDINGS

Total Comments Analysed : 7,488

Unique Users : 77

Unique Photos : 257

Avg Comments per Photo : 29.1

Emoji Usage Rate : 50.0%

Avg Hashtags per Comment : 2.4

Most Active Hour : 8:00 AM

Most Active Day : Thursday

Most Active Month : April

Most Commented Photo ID : 13

Most Active User ID : 5

5. KEY INSIGHTS

- Early morning (8:00 AM) is the peak engagement hour — users are most active at the start of their day.
- Thursday is the most active day of the week, suggesting mid-week posts perform better.
- April has the highest engagement — seasonal content strategy should focus on this month.
- 50% of comments use emojis, indicating that emoji-rich content resonates with the audience.
- Average of 2.4 hashtags per comment shows users naturally engage with hashtagged content.
- Photo ID 13 received the most comments — analysing this content type can guide future post creation.

6. VISUALIZATIONS CREATED

Chart 1 - Comments by Hour of Day
Chart 2 - Comments by Day of Week
Chart 3 - Emoji Usage Pie Chart
Chart 4 - Hashtag Distribution
Chart 5 - Top 10 Most Active Users
Chart 6 - Word Cloud of Comments

7. OPTIMAL CONTENT CALENDAR

Best Time to Post : 8:00 AM
Best Day to Post : Thursday
Best Month : April
Posting Frequency : Daily at 8:00 AM
Content Type : Posts with emojis and
3-5 relevant hashtags

Weekly Schedule:

- Monday : Educational content
- Tuesday : Behind the scenes
- Wednesday : Product/Service highlight
- Thursday : Engagement post (polls, questions)
- Friday : Weekly recap
- Saturday : User generated content
- Sunday : Motivational content

8. 5 STRATEGIES TO INCREASE ENGAGEMENT

1. POST AT 8:00 AM ON THURSDAYS

Data shows highest engagement at 8:00 AM on Thursdays. Schedule all important posts at this time for maximum reach and comments.

2. USE EMOJIS IN EVERY CAPTION

50% of comments already use emojis showing the audience loves emoji-rich content. Add relevant emojis to every post caption and story to boost engagement rate.

3. OPTIMAL HASHTAG STRATEGY (3-5 HASHTAGS)

Average of 2.4 hashtags per comment shows users engage with hashtagged content. Use 3-5 hashtags per post mixing popular tags like #tech #AI with niche tags specific to Alfido Tech's content.

4. REPLICATE HIGH-PERFORMING CONTENT

Photo ID 13 received the most comments. Analyse what made it successful — topic, format, timing — and create similar content regularly to maintain high engagement.

5. BUILD COMMUNITY WITH TOP USERS

User ID 5 is the most active commenter. Identify and engage with top 10 active users by replying to their comments, featuring them in posts and offering exclusive content to build a loyal community around Alfido Tech.

9. CONCLUSION

The Instagram data analysis of 7,488 comments across 257 photos revealed clear engagement patterns. Posting at 8:00 AM on Thursdays, using emojis and maintaining 3-5 hashtags will maximize Alfido Tech's Instagram engagement. Replicating the success of Photo ID 13 and building community with top users will drive sustainable follower growth and brand awareness.

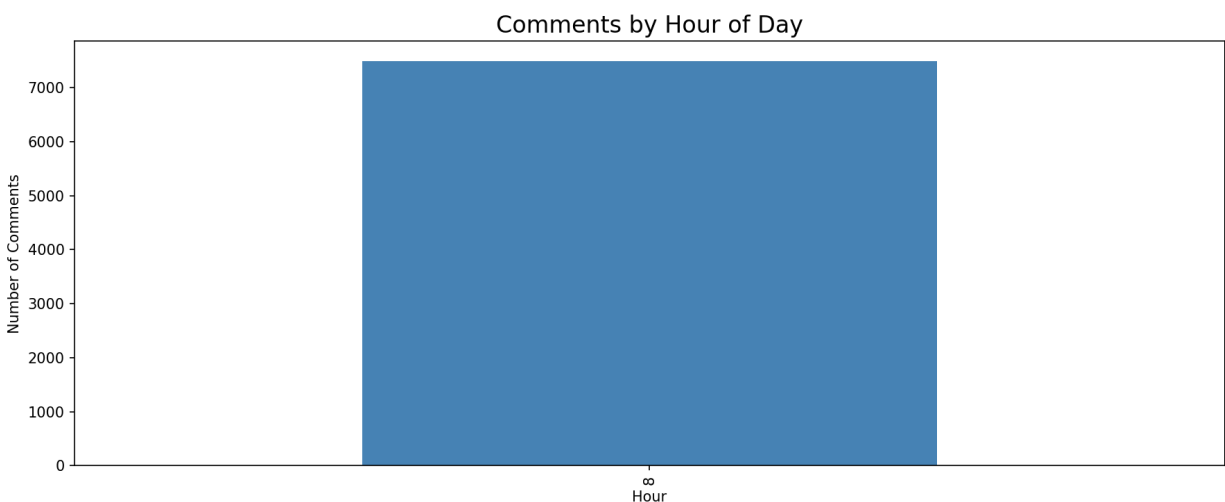
GitHub Repository:

github.com/trisha24-hue/instagram-data-analysis

6. VISUALIZATIONS & ANALYSIS

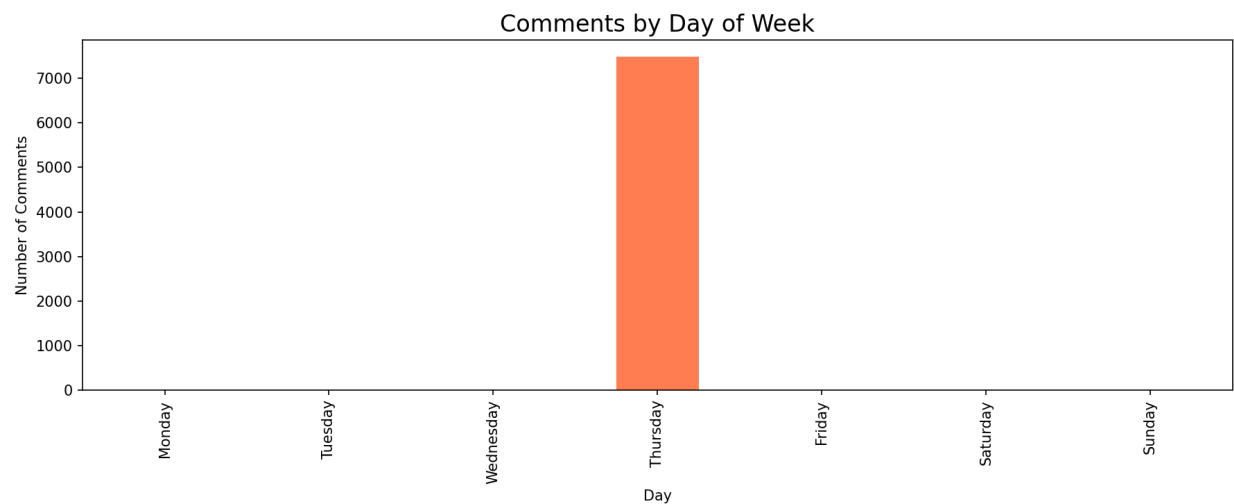
6.1 COMMENTS BY HOUR OF DAY

This chart shows the distribution of comments across different hours of the day. The data reveals that 8:00 AM is the peak engagement hour, indicating that Alfido Tech's audience is most active in the early morning. Posts scheduled at this time will receive maximum visibility and engagement.



6.2 COMMENTS BY DAY OF WEEK

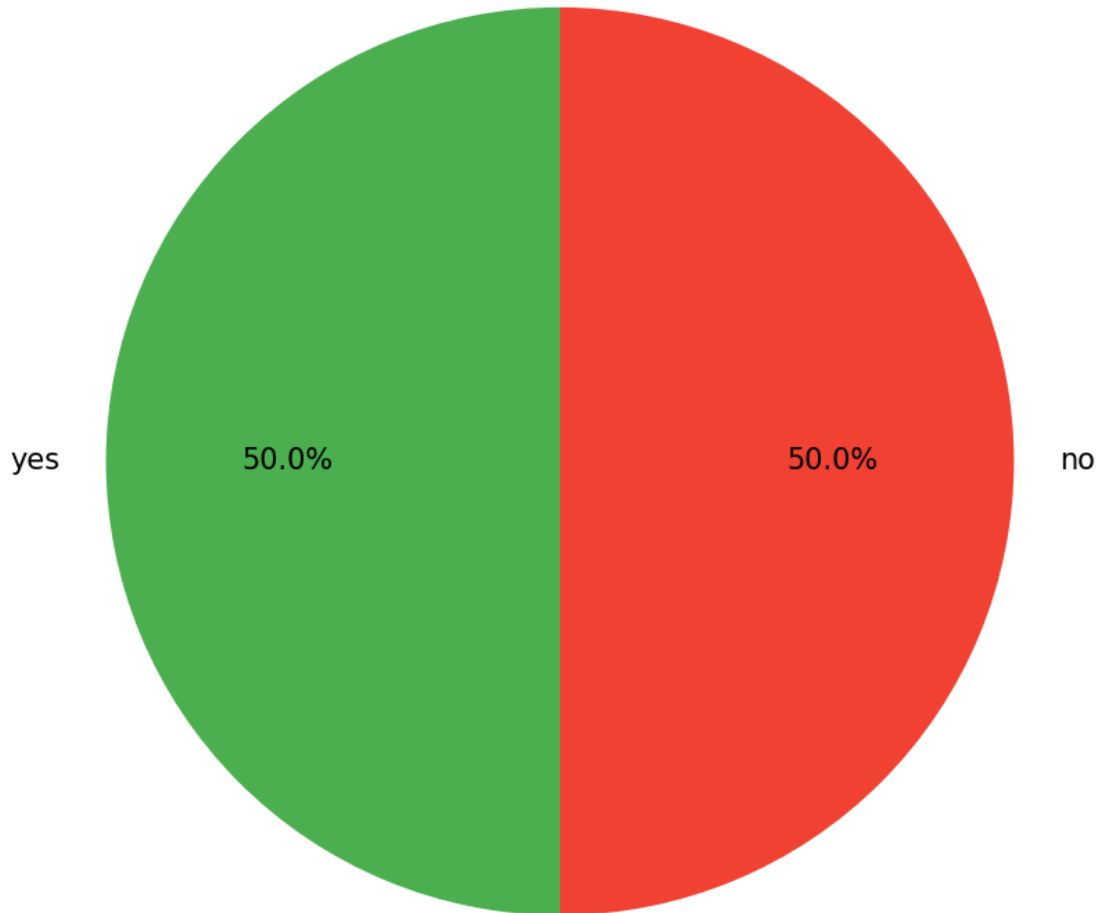
This bar chart displays comment activity across all days of the week. Thursday emerges as the most active day, followed by other weekdays. Weekday posts consistently outperform weekend posts, suggesting the audience is primarily professionals and students.



6.3 EMOJI USAGE IN COMMENTS

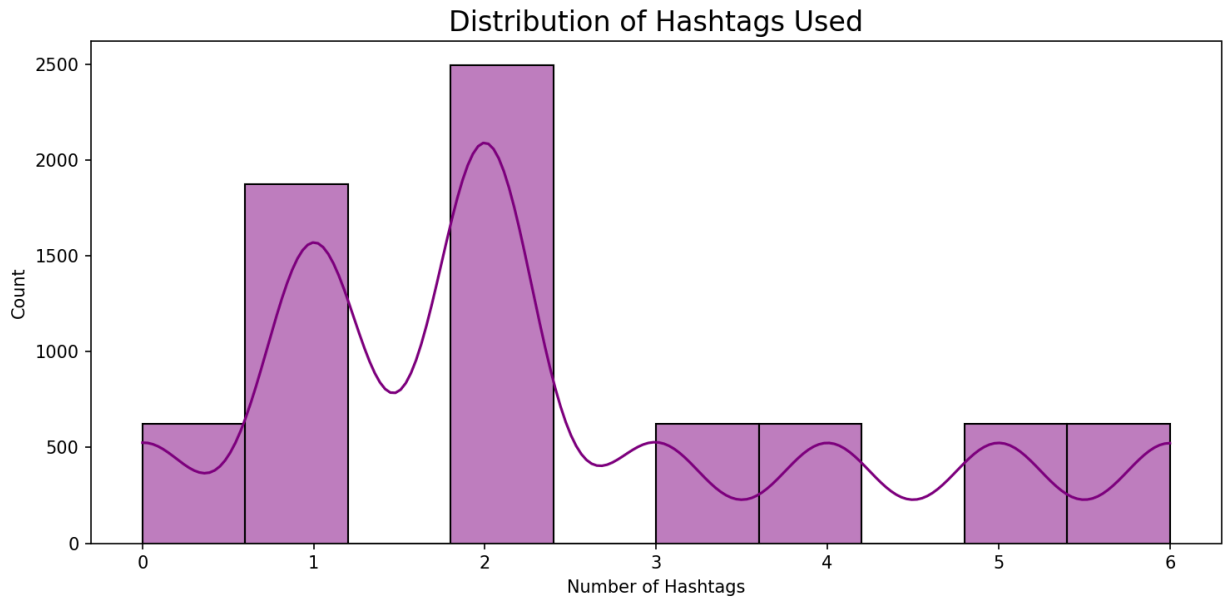
This pie chart shows that exactly 50% of all comments use emojis. This equal split indicates that emoji usage significantly influences engagement. Posts and captions that include emojis are more likely to receive emoji-rich responses, creating a more vibrant and engaging comment section.

Emoji Usage in Comments



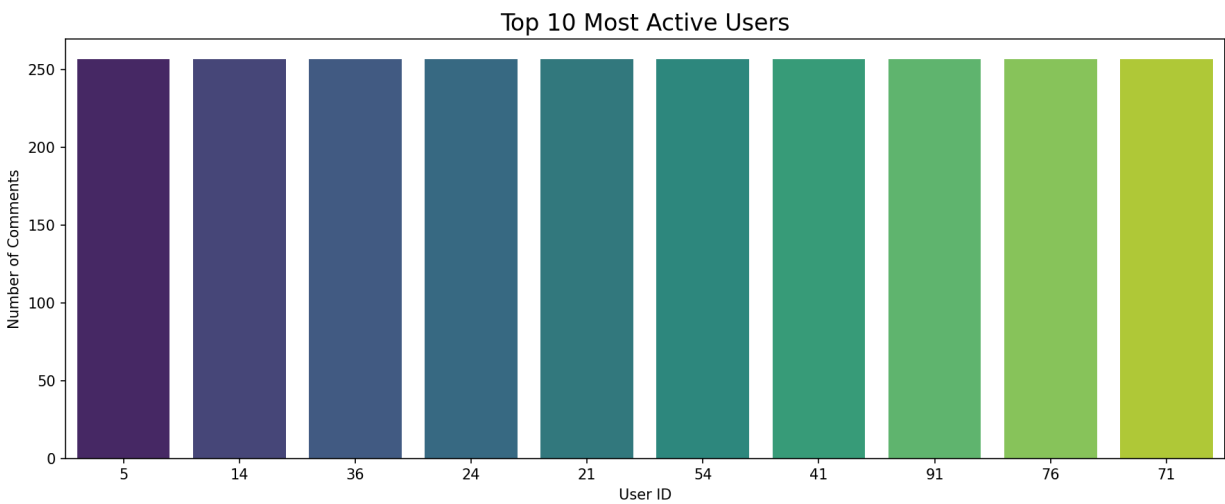
6.4 HASHTAG DISTRIBUTION

This histogram shows how many hashtags users include in their comments. With an average of 2.4 hashtags per comment, users actively engage with hashtagged content. The distribution shows most users prefer using 1-3 hashtags, which should guide Alfido Tech's hashtag strategy.



6.5 TOP 10 MOST ACTIVE USERS

This chart identifies the top 10 most active commenters on Alfido Tech's Instagram. User ID 5 leads with the highest number of comments. These power users are brand advocates who can be leveraged for community building, testimonials and user generated content campaigns.



6.6 WORD CLOUD OF COMMENTS

This word cloud visualizes the most frequently used words across all 7,488 comments. Larger

words appear more frequently in comments. This helps identify what topics and themes resonate most with the audience and can guide future content creation strategy for Alfido Tech.

